

STUDENT PERFORMANCE PREDICTOR

User Manual

Complete guide to installing, setting up, and using the app effectively

Version 1.0 | May 2026

1 Introduction

Student Performance Predictor is an Android mobile application that helps students assess and improve their academic performance. By entering key study-related parameters, the app calculates a predicted score and grade, identifies weaknesses, and provides personalized AI-powered advice.

The app also features a unique **Smart App Lock**: if your predicted performance falls below 50%, the app locks for 1 hour to encourage focused study time. This manual will guide you through every feature of the application.

Tip: Be honest when entering your parameters — the more accurate your inputs, the more useful and accurate your performance prediction will be.

2 System Requirements

REQUIREMENT	MINIMUM	RECOMMENDED
Android Version	Android 8.0 (API 26)	Android 12.0+
RAM	2 GB	4 GB or more
Storage	50 MB free	200 MB free
Internet	Required for login & AI insights	Stable Wi-Fi or mobile data
Screen	5-inch display	6-inch or larger

3 Installation Guide

Method A: Install from APK

1. Download the `StudentPerformancePredictor.apk` file from the repository's `apk/` folder.
2. Transfer the APK to your Android device (via USB, email, or cloud storage).
3. On your Android device, open **Settings** → **Security** (or **Settings** → **Apps** → **Special App Access**).
4. Enable "**Install from Unknown Sources**" for your file manager app.
5. Open the APK file using your file manager and tap **Install**.
6. Once installed, open **Student Performance Predictor** from your app drawer.

Note: You may see a "Play Protect" warning. This is normal for APKs installed outside the Play Store. Tap "**Install Anyway**" to proceed.

4 Getting Started

4.1 Creating an Account

1. Open the app. You will see the **Welcome Back** login screen.
2. Tap "**Don't have an account? Sign Up**" at the bottom.
3. Enter your **email address** and choose a strong **password** (minimum 6 characters).
4. Tap **Sign Up**. Your account will be created and you will be logged in automatically.

4.2 Logging In

1. Open the app and enter your registered **email address**.
2. Enter your **password** (tap the eye icon  to show/hide).
3. Tap **LOGIN**. You will be taken to the main dashboard.

Forgot your password? Contact the app developer or use Firebase's password reset feature if integrated.

5 **Using the Main Dashboard**

The main screen is titled "**Student Performance Predictor**" and contains the **INPUT PARAMETERS** panel with 7 interactive sliders. Adjust each slider to accurately reflect your current academic situation.

PARAMETER	RANGE	DESCRIPTION
 Daily Study Hours	0 – 12 hours	Average hours you study per day
 Attendance Rate	0 – 100%	Percentage of classes you attend
 Assignment Completion	0 – 100%	Percentage of assignments you complete and submit
 Class Participation	0 – 100%	How actively you participate in class discussions
 Sleep Hours/Night	0 – 12 hours	Average hours of sleep you get per night
 Previous Grade	0 – 100%	Your grade/score from the previous term or exam
 Stress Level	0 – 100%	Your current level of academic stress (higher = more stressed)

Once you have adjusted all sliders, tap the " **ANALYZE PERFORMANCE**" button to generate your results.

6 Understanding Your Analysis Results

6.1 Predicted Score & Grade

The top card displays your **Predicted Score** (0–100%) and the corresponding letter grade:

SCORE RANGE	GRADE	PERFORMANCE LEVEL
85% – 100%	A	Excellent Performance
70% – 84%	B	Good Performance
55% – 69%	C	Average Performance
40% – 54%	D	Below Average
0% – 39%	F	Failing — Needs Immediate Improvement

6.2 Skill Radar Chart

The **Skill Radar** (spider chart) visualizes all 7 of your input parameters on a hexagonal grid. Each axis represents one parameter. A larger, more balanced shape indicates well-rounded academic habits. A small or lopsided shape highlights specific areas needing attention.

6.3 Weakness Analysis

This section automatically identifies your weakest academic areas based on the parameters you entered. If all parameters are above the threshold, you will see: " *No critical weaknesses detected! Keep it up!*"

6.4 AI Advisor Insights

Powered by an AI model, this section provides personalized academic recommendations based on your specific profile. If the AI service is unavailable, you will see: "*Could not parse AI response. Please try again.*" — in this case, check your internet connection and retry.

7 Smart App Lock Feature

If your predicted performance score is **below 50%**, the app will activate the **Smart App Lock** feature automatically.

1

Lock Triggered

A lock screen appears with a red padlock icon and the message: "Your performance is below 50%. Take some time to focus on your studies."

2

Countdown Timer

A 1-hour countdown timer (00:59:55) is displayed. You cannot access the main app features during this time.

3

Focus Time

Use this 1 hour to study, review notes, or address the weaknesses identified in your analysis.

4

Lock Expires

Once the timer reaches 00:00:00, the app unlocks automatically. You can then re-analyze with improved parameters.

Important: The App Lock is a motivational tool, not a parental control. Closing and reopening the app will not bypass the timer — the lock persists until the countdown completes.

8 Tips for Best Results

- Enter your parameters **honestly** — inflating values will give an inaccurate prediction
- Use the Skill Radar to identify your **weakest axis** and focus your improvement there
- Aim for at least **75% attendance** and **70% assignment completion** for a B grade
- Maintain **7–9 hours of sleep** per night for optimal cognitive performance
- Re-run the analysis **weekly** to track your progress over time
- Pay attention to the **AI Advisor Insights** — they are tailored specifically to your profile

9 Troubleshooting

PROBLEM	POSSIBLE CAUSE	SOLUTION
Cannot log in	Wrong email or password	Double-check credentials; ensure Caps Lock is off
AI Advisor shows error	No internet connection or API limit	Check internet connection and try again
App stuck on lock screen	Timer still running	Wait for the countdown to complete
App crashes on launch	Outdated Android version	Update Android to 8.0+ or reinstall the APK
Sliders not responding	Touch calibration issue	Restart the app or clean the screen
Analysis result not showing	Internet timeout	Check connection and tap Analyze again

10 Contact & Support

For technical support, bug reports, or feature requests, please contact the developer:

- **Developer:** UMAR BALOC
- **Email:** ub1529581@gmail.com
- **GitHub Issues:** github.com/Umer-7585
- **University:** University of Layyah , Computer Science